

Enterprise Agent Evolution

From Copilot to Multi-Agent · From Q&A to Harness

A panoramic study on the architectural evolution of enterprise AI Agents

Session Overview

Three Eras · Three Paradigms · One Evolution Path

01 The Starting Point

Why do we need Agents? How AI reshapes enterprise productivity

03 Single-Agent Era

Workflow-driven · Automated task closure

05 Comparison & Insights

Three-generation capability comparison · Enterprise adoption path

02 Copilot Era

Q&A assistant paradigm · Human-in-the-loop

04 Multi-Agent Era

Harness orchestration · Multi-agent collaboration

06 Future Outlook

From tools to employees · Self-evolving organization

Why do enterprises need Agents?

AI is no longer just answering questions — it is completing work



The value of AI is shifting from “generating content” to “completing business operations” .

What enterprises care about is never model size, but: Who executes the task? Who owns the process? Who is accountable for results?

01 Productivity Demand

Knowledge workers expect AI to take over repetitive, cross-system multi-step tasks, not just answer questions.

02 Data & Context

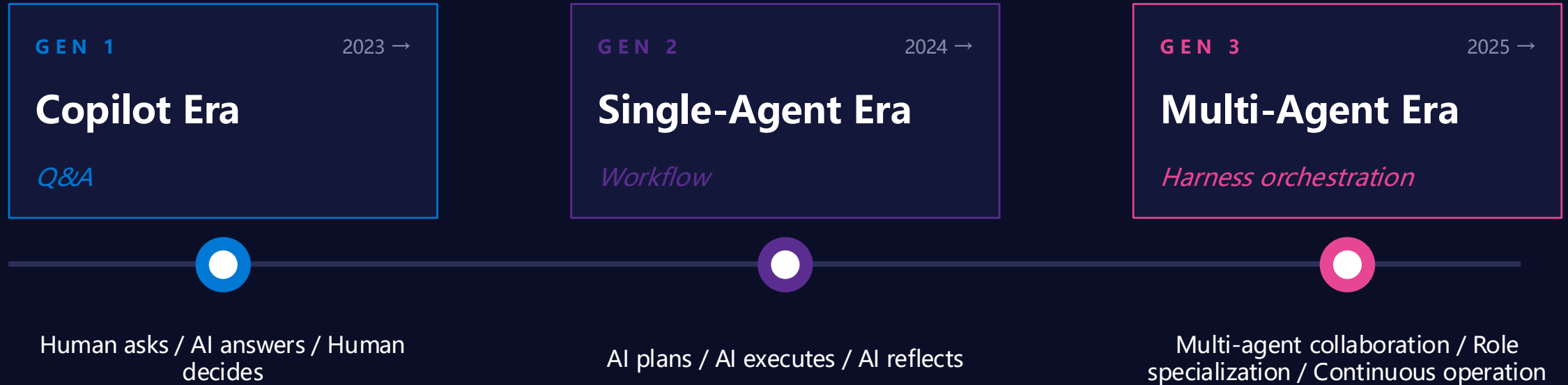
Enterprise data is ready (M365, ERP, CRM); we need execution agents that “understand” this context.

03 Organizational Complexity

A single model cannot cover end-to-end processes; we need a “digital workforce network” with multi-role collaboration.

Three Eras • Three Paradigms

From “assisting humans” to “replacing humans” to “organizing humans”



Evolution path: Automation ↑ · Human intervention ↓ · Task complexity ↑ · Architecture scope ↑

PART 01

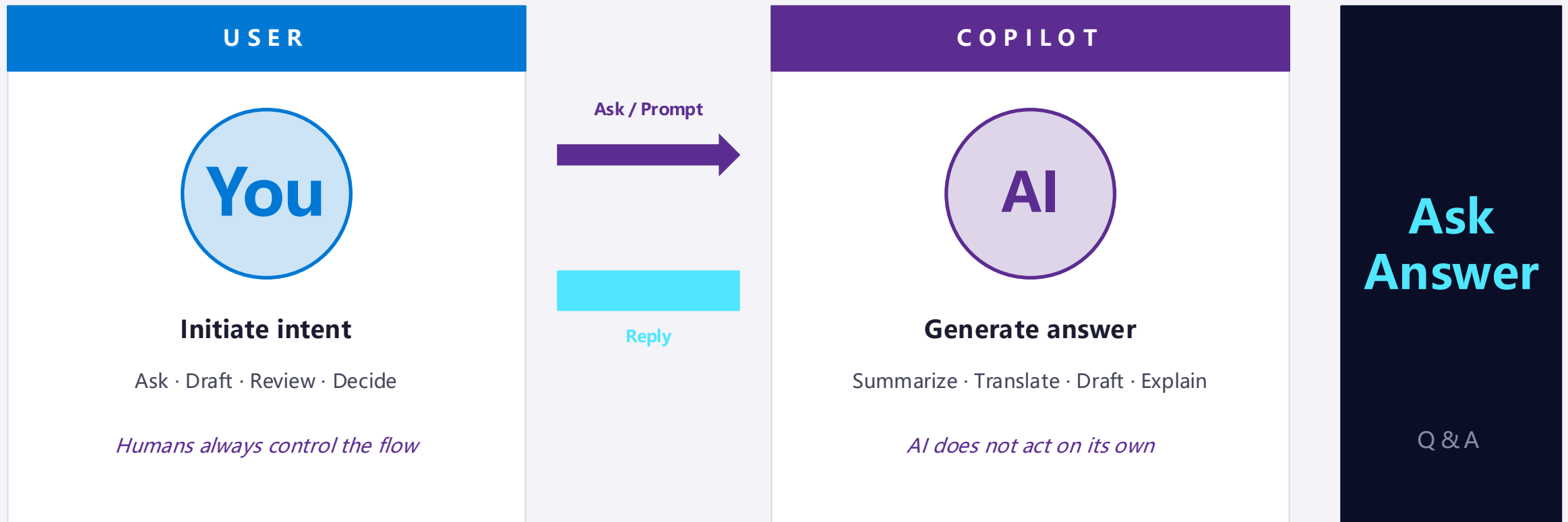
Copilot Era

Q&A · Human-in-the-loop · AI as Copilot

When we say Copilot, AI is still in an “assisting” role.
It excels at content generation and single-point answers, but humans still hold the wheel.

Copilot: AI as the “Co-pilot”

User requests / Copilot suggests / User decides



The Boundaries of Q&A Capabilities

What Copilot does: Content generation + Information lookup



Writing emails

Draft professional emails from context

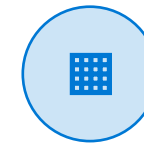
PROMPT → REPLY



Meeting notes

Summarize meeting key points & to-dos

PROMPT → REPLY



Data queries

Natural-language query over data tables

PROMPT → REPLY



Document summary

Extract key info from long documents

PROMPT → REPLY



Code completion

Generate code snippets for developers

PROMPT → REPLY



Translation & polishing

Cross-language rewriting and expression refinement

PROMPT → REPLY

Essence: AI is a “tool” , not an “executor” . Every step requires humans to initiate, confirm, and accept.

Why Copilot is not enough

When tasks span steps, systems, and time, single-point dialogue hits a ceiling

× Stateless

Conversation ends → context lost; cannot persist across tasks

× No Action

Produces only suggestions — no system actions, no API calls, no DB writes

× No Planning

Lacks task decomposition and goal-driven multi-step reasoning

× Human-dependent

Every step needs user input; hard to run asynchronously

Real user experience

Me:

Compile all transactions for Customer A and send to the Sales Director

AI:

Here is the draft content... (you still need to click send)

Me:

Process Customer B as well, grouped by month

AI:

Please re-supply Customer B's data — context limit exceeded.

→ *Pain point: every time you re-feed data and re-click send*

PART 02

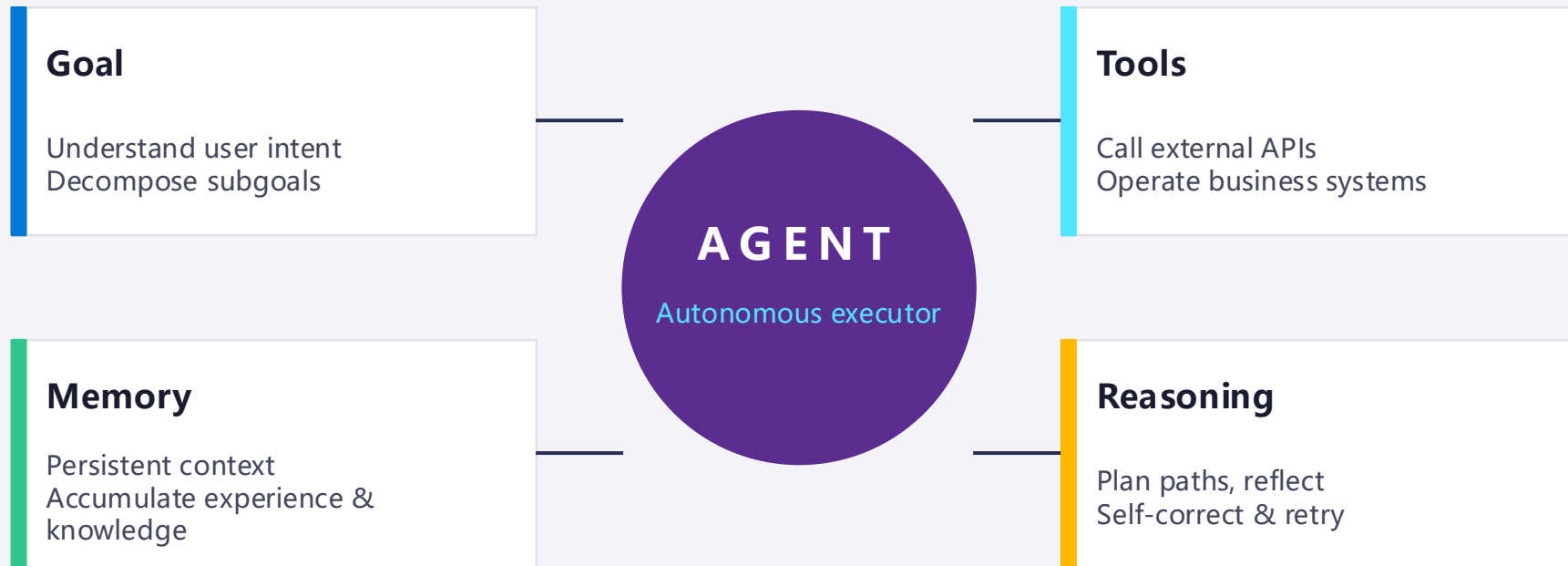
Single-Agent Era

From "Q&A" to "Workflow" · AI takes over multi-step tasks

An Agent has goals, memory, tools, and reflection capabilities.
It no longer just answers — it plans, invokes, executes, and judges — a complete task loop.

What is an Agent?

Goal + Memory + Tools + Reasoning = an agent that can autonomously complete tasks



Agent ≠ a smarter Copilot

It is a digital executor with "action authority" and "closed-loop capability" .

From “Q&A” to “Workflow”

The Agent strings multi-step actions into one executable pipeline

BEFORE · Copilot Q&A

User ▶ Look up new contracts signed this week

AI ▶ (Provides SQL)

User ▶ (Manually execute → Copy results)

User ▶ Now turn it into a chart

AI ▶ (Outputs suggested steps)

User ▶ (Manually execute → Send email)

AFTER · Agent Workflow

1. Receive goal

“Compile this week's new contracts and notify the Sales Director”

2. Plan tasks

Decompose: Query → Aggregate → Chart → Email

3. Call tools

CRM API · BI engine · Email system

4. Reflect & verify

Check data integrity, retry if needed

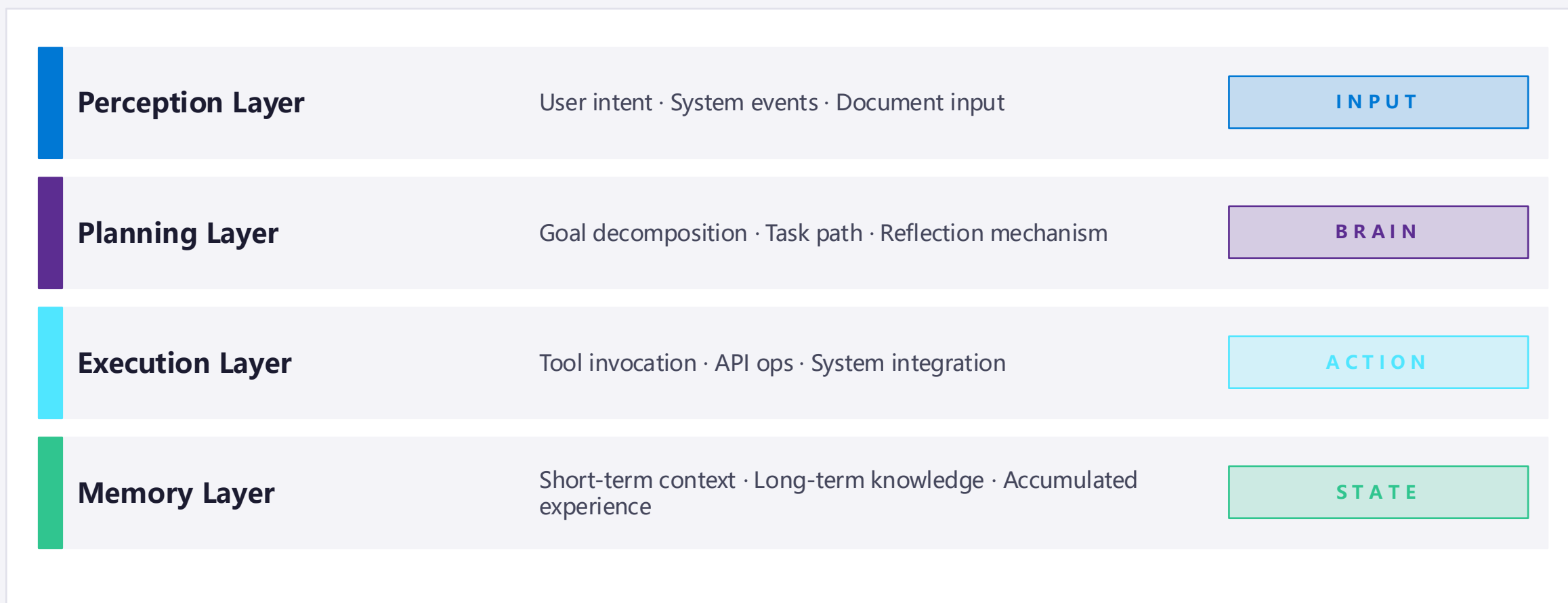
5. Complete delivery

Email sent, report execution result

Humans no longer drive each step — they only “set goals” and “confirm results”

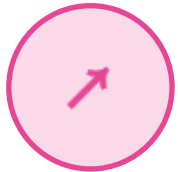
Standard Single-Agent Architecture

Plan → Tool Use → Memory → Reflect loop engine



The Limits of a Single Agent

One Agent doing everything will eventually hit a “capability ceiling”



Context Explosion

The longer the task, the more tokens. One Agent cannot hold all domain contexts at once.



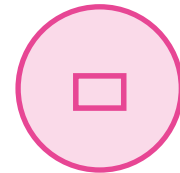
Role Conflict

Writing emails, reviewing contracts, running finance — a single prompt cannot carry multiple specialties.



Debugging Black Box

When a long chain fails, hard to tell if planning, tools, or memory is at fault.



Insufficient Elasticity

Cannot parallelize, cannot hot-swap components, hard to scale business complexity horizontally.

Conclusion: A single Agent still “fights alone” . To solve organizational problems, we must move from “individual” to “team” .

PART 03

Multi-Agent Era

From "Workflow" to "Harness" · Agents form a digital team

A Harness is like an org chart: each Agent is a role, each with its own duty, uniformly scheduled, observed, and aligned to goals.

Multi-Agent: Digital Workforce Network

Specialized Agents collaborate to complete end-to-end business



Harness: The “Operating System” for Agents

Not more Agents — better orchestration of Agents

H A R N E S S

A runtime that constrains, schedules, and amplifies the capabilities of Agents.

It defines:

- How to route tasks to the right Agent
- How to share state, memory, and tools
- How to observe, replay, and correct every execution
- How to expose secure, controlled interfaces externally

Orchestration

Decides who does what and when. Goal-based auto-routing and orchestration.

Shared Context

Memory, files, and variables shared across Agents, avoiding redundant dialogue and context loss.

Observability

Every step is traceable, replayable, and auditable, turning the black box into a white box.

Governance

Permissions, quotas, and compliance policies are uniformly injected, giving enterprises the confidence to deploy Agents in production.

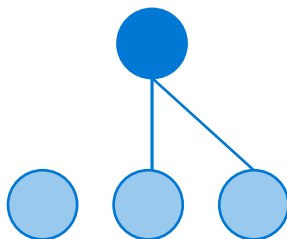
How do Agents “cooperate” ?

Four proven collaboration topologies

Hierarchical

A “Manager Agent” assigns tasks to subordinate Agents

Typical: complex business flows · decision authority needed



Pipeline

Agents connect sequentially; previous output = next input

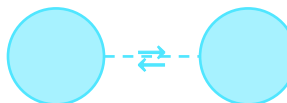
Typical: data processing · content production · approval chains



Debate

Multiple Agents offer different views and challenge each other

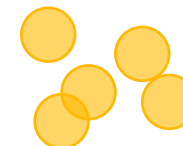
Typical: strategy evaluation · investment decisions · high-risk reviews



Market

Tasks broadcast; Agents bid / claim autonomously

Typical: elastic scaling · heterogeneous skills · real-time response



Capability Comparison Across Three Eras

One table to connect the entire evolution

Dimension	Copilot Era	Single-Agent Era	Multi-Agent Era
Interaction Mode	Q&A	Workflow	Harness orchestration
Proactivity	Passive response	Goal-driven	Always on
Execution Authority	Suggests	Can invoke tools	Collaborative decisions
Context	Conversation-level	Task-level + Memory	Organization-level + Shared bus
Runtime Scope	Single-point answer	Single-Agent loop	Multi-Agent collaboration
Human Role	Driver	Project Manager	Org Designer

Trend: from “Ask AI” → “Let AI do” → “Let a team of AIs do it together”

What does it mean for the enterprise?

Not replacing tools — reshaping how the organization operates

10×

Task automation coverage

From single-point assistance → end-to-end processes

70%

Repetitive work reduced

Agents take over the collaboration glue layer

24/7

Always on

Harness keeps Agents on duty

N → 1

Unified tool entry

Users face Agents, not systems

Finance · Risk & Compliance Review

Audit + Compliance + Report Agents — materials to conclusions, closed-loop.

Manufacturing · Supply Chain

Procurement + Logistics + Risk Agents collaborate in real time, responding to anomalies cross-system.

Retail · Customer Operations

Service + Marketing + Data Agents share the customer journey.

ONE LINE TO REMEMBER

**Copilot brings AI into the enterprise,
Agents let AI complete the work,
Harness makes AI an organization.**

Thank you.